

JUNBO NIU

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Education

Peking University

Ph.D. Candidate, **EECS**

Beijing, China

2025.09 – 2030.06

- Focus on **LLM**(UT Modeling Diffusion-LM && Loop-LM) and **Computer Use Agent**.

Beihang University

B.S, **EE**

Beijing, China

2021.09 – 2025.06

- Focus on **Video World Model** and **Multimodal Foundation Model**.
- GPA: 3.85/4.00, Rank: 1/203 (Top 1%)

Selected Publications (* denotes equal contribution)

(Co-) First Author in **Multimodal Foundation Model**

Native Visual Understanding: Resolving Resolution Dilemmas in Vision-Language Models (CVPR 2026) 📄

- Identified critical resolution dilemmas in Vision-Language Models and their impact on fine-grained perception.
- Proposed NativeRes-LLaVA, enabling native-resolution visual encoding and significantly improving detail preservation and performance.
- **Junbo Niu***, Yuanhong Zheng*, Ziyang Miao, *et al.*

MinerU-Diffusion: Rethinking Document OCR as Inverse Rendering via Diffusion Decoding (ECCV in Process) 📄

- Reformulated OCR as an inverse rendering problem with a diffusion-based decoding paradigm for robust document understanding.
- MinerU-Diffusion enables flexible accuracy–throughput trade-offs, achieving up to $3.26\times$ TPS vs AR (e.g., $2.12\times$ at 99.9% accuracy, $3.01\times$ at 98.8%).
- Hejun Dong*, **Junbo Niu***[‡] (Project Lead), Bin Wang, *et al.*

MinerU2.5-Pro: Pushing the Limits of Data-Centric Document Parsing at Scale (Neurips 2026 in Process) 📄

- Demonstrated that data engineering, rather than model scaling, is the primary driver for high-performance document parsing.
- Introduced a data-centric optimization paradigm with large-scale data curation and training strategies, significantly improving robustness and accuracy.
- Bin Wang*, **Junbo Niu***, *et al.*

MinerU2.5: Decoupled Vision-Language Model for Efficient High-Resolution Document Parsing (ACL2026 Oral) 📄

- Proposed a decoupled vision-language architecture for efficient high-resolution document parsing.
- Achieved strong performance on complex layout understanding and structured document parsing tasks.
- **Junbo Niu***, Zhenyu Liu*, Zhihao Gu*, *et al.*

MinerU2.0: End-to-End Vision-Language Model for High-Speed and Precision Document Parsing (Model Card)

- Developed an end-to-end multimodal document parsing model with fewer than 1B parameters, outperforming large-scale (72B) VLMs in accuracy.
- Supports multilingual OCR, handwriting recognition, layout analysis, table parsing, and formula recognition.
- Achieves over 10,000 tokens/s throughput with optimized inference (sglang on RTX 4090).
- **Junbo Niu***, Zhenyu Liu*, Zhihao Gu*, *et al.*

(Co-) First Author in **Video World Model**

OVOBench: How Far is Your Video-LLMs from Real-World Online Video Understanding? (CVPR 2025). 📄

- Proposed a benchmark for evaluating real-world online video understanding in Video-LLMs, bridging the gap between offline evaluation and practical streaming scenarios.
- Covered streaming/online video reasoning, temporal understanding, and real-time evaluation protocols.
- **Junbo Niu***, Yifei Li*, Ziyang Miao, *et al.*

Seed 2.0: Towards Intelligence Frontier for Real-World Complexity (Model Card)

- A production-oriented multimodal agent model series with improved reasoning, instruction following, and long-horizon task capabilities.

- Enhanced performance in visual understanding, efficiency, and real-world deployment scenarios.
- ByteDance Seed Team

Seed 1.8: Towards Generalized Real-World Agency (Model Card)

- A generalized agentic multimodal model integrating search, coding, and GUI interaction capabilities.
- Designed for complex real-world workflows and autonomous task execution.
- ByteDance Seed Team

Projects

MinerU (60.0 K GitHub stars, Core Contributor in Data/Modeling/Training/Infra.)  2024.09 – 2026.05

- **Transforms complex web Content like HTML and documents like PDFs into LLM markdown/JSON for your Agentic workflows.**
- Engineered a **unified, end-to-end model for general OCR tasks**, significantly boosting performance and operational efficiency.
- Designed and implemented the **core model training framework**, actively exploring and integrating novel model architectures.
- Streamlined data pipelines and **the optimization of the model architecture’s training process**.

DataFlow (3.2K GitHub stars, Contributor.)  2024.06 – now

- **Generate, Clean, and Prepare LLM Data, All-in-One.**
- Pioneered the DataCoT method for generating high-quality visual instruction data, significantly reducing costs by leveraging open-source large language models.
- Design the filtering, processing, and generation of large-scale video datasets to enhance multimodal learning quality and efficiency.
- Designed and implemented the foundational framework for scalable video data processing and generation pipelines.

Experience

ByteDance Seed 2025.09 – 2026.04

Research Intern at Multimodal Interaction & World Model (Seed1.6/1.8/2.0)

- Focus on **Video World Model**
- **Video SpatioTemporal World Modeling**
- **Video Reasoning Via SpatioTemporal Modeling**
- **Video Agentic Reasoning Via VideoCut**

Shanghai AI Lab 2024.06 – 2025.09

Research Intern at Large Model Center (MinerU1.0/2.0/2.5/2.5-Pro/Diffusion/HTML)

- Focus on **Multimodal Foundation Model**
- **Video-LLMs in long-context understanding**
- **Web Content Understanding and Extraction**
- **High-Precision OCR Model Modeling and Data Preparation System**

Honours and Awards

Outstanding student of Beihang University	09/2022
Merit Student of Beihang University	12/2022
Star of “Yu-Yuan”, Top 2 of 600+ ,The highest honor of the department.	12/2024
National Scholarship, (Ministry of Education, People’s Republic of China)	12/2024

Skills

- Languages:** Familiar with Python and C++/C
- Tools:**Pytorch, Git, Linux
- Language:**English, Mandarin - Native speaker
- Miscellaneous:** Basketball (PB: Departmental champion), Outdoor sport